particle. A tiny object; one of the building blocks of matter (such as an electron, proton, photon, or graviton).

perihelion. The location, on a planet's orbit around the Sun, at which it is closest to the Sun.

perihelion shift of Mercury. The tiny failure of Mercury's elliptical orbit to close on itself, which results in its perihelion shifting in position each time Mercury passes through the perihelion.

perturbation. A small distortion (from its normal shape) of an object or of the spacetime curvature around an object.

perturbation methods. Methods of analyzing, mathematically, the behaviors of small perturbations of an object, for example, a black hole.

photon. A particle of light or of any other type of electromagnetic radiation (radio, microwave, infrared, ultraviolet, X-ray, gamma ray); the particle which, according to wave/particle duality, is associated with electromagnetic waves.

piezoelectric crystal. A crystal that produces a voltage when squeezed or stretched.

Planck's constant. A fundamental constant, denoted $\hbar$, that enters into the laws of quantum mechanics; the ratio of the energy of a photon to its angular frequency (that is, to 2π times its frequency); 1.055×10^{-27} erg-second.

Planck-Wheeler length, area, and time. Quantities associated with the laws of quantum gravity. The Planck—Wheeler length, $\sqrt{G\hbar/c^3} = 1.62 \times 10^{-33}$ centimeter, is the length scale below which space as we know it ceases to exist and becomes quantum foam. The Planck—Wheeler time (lie times the Planck-Wheeler length or about 10^{-43} second) is the shortest time interval that can exist; if two events are separated by less than this, one cannot say which comes before and which after. The Planck–Wheeler area (the square of the Planck–Wheeler length, that is, 2.61×10^{-66} square centimeter) plays a key role in black-hole entropy. In the above formulas, $G = 6.670 \times 10^{-8}$ dyne-centimeter^2/gram^2 is Newton's gravitation constant, $\hbar = 1.055 \times$